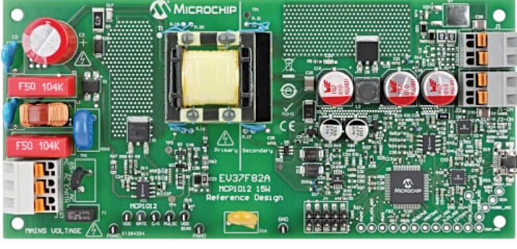
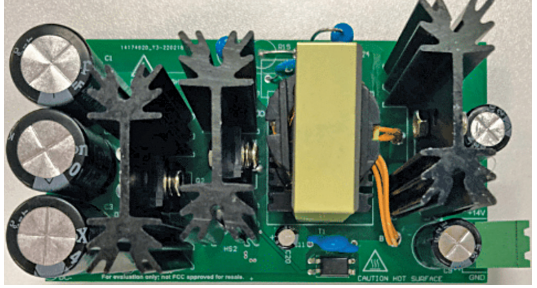


Interesting Reference Designs Of ISOLATED POWER SUPPLIES

Title of Reference Design	15W Isolated Power Supply	Flyback Isolated Power Supply
Image		
Key Attractions	The reference design showcases a unique method of secondary side control, featuring high-voltage startup capabilities, programmable oscillators, and robust safety features for sensitive applications.	The reference design delivers a +14V, 56W output and features high precision in voltage regulation and superior EMI performance across a wide 350V to 1000V input range.
Highlights	The EV37F82A from Microchip is an isolated power supply reference design that features the MCP1012 15W, which demonstrates the startup capability of a 15W AC-DC flyback converter by the MCP1012 on the primary side, subsequently controlled by a microcontroller on the secondary side. This setup evaluates a novel method of isolated secondary side control using the Microchip MCP1012 along with a Microchip patented integrated magnetic winding transformer manufactured by Würth Electronic. An isolated power supply, such as this, ensures there is no electrical connection between its input and output. It is crucial for applications like medical equipment and industrial control systems, where isolation prevents signal interference. The reference design is equipped with a high-voltage startup capability of 500V continuous and 700V transient with a series 10k resistor, a programmable low-frequency oscillator for setting startup power, and the ability to receive external PWM commands via an isolator. It includes multiple safety features such as undervoltage and overvoltage lockout protections, cycle-by-cycle current limiting, and thermal shutdown for overtemperature conditions.	The PMP41009 reference design by Texas Instruments is an isolated power supply featuring a +14V, 56W output, utilising a quasi-resonant flyback topology capable of operating across a wide DC input range of 350V to 1000V. At its heart, cascode MOSFETs and the UCC28740-Q1 flyback controller deliver line and load regulation within $\pm 1\%$ precision, achieving a peak efficiency of 84.6% with a 1000V input at full load. This design is particularly suited for auxiliary power supplies, boasting high-precision output voltage and superior EMI performance. The efficiency graph illustrates the system's performance across varying output currents for inputs of 350V and 1000V, demonstrating high efficiency that begins around 0.83 and peaks over 0.85 at lower inputs, while showing gradual improvement and stabilisation around 0.81 at the higher input level as output current increases.
Applications	The reference design is suitable for medical equipment, scientific instruments, and industrial control systems.	Auxiliary power supplies are a key application area for this design.
OEM Brand	Microchip	Texas Instruments (TI)
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-15-w-isolated-power-supply	https://www.electronicsforu.com/electronics-projects/flyback-isolated-power-supply-reference-design